

Def.

A subset $S \subseteq \mathbb{R}^3$ is called a Regular Surface if:

for each $p \in S$, there are open neighbor $V \subset \mathbb{R}^3$ containing p and open $U \subset \mathbb{R}^2$ with

$X: U \rightarrow V \cap S$ which satisfies

- 1). X is C^∞ .
- 2). X is Homeomorphism
- 3). dX_q is 1-1 for each $q \in U$.

Thm. If $f: U \rightarrow \mathbb{R}$ is differentiable on an open $U \subseteq \mathbb{R}^2$, then $S = \{(x, y, f(x, y)) \mid (x, y) \in U\}$ is regular surface.

Proof. Enough to show that $X: U \rightarrow S: (u, v) \mapsto (u, v, f(u, v))$ is Parametrization.

For each $q \in U$, $dX_q = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ f_x & f_y \end{pmatrix}$ has rank 2 clearly, so 1-1.

X is C^∞ since f is differentiable. Moreover, X is clearly 1-1, so X^{-1} is also differentiable, continuous, so proved $\square$

Def. Let $U \subseteq \mathbb{R}^n$ be an open set, and let $F: U \rightarrow \mathbb{R}^m$ be a mapping. A point $p \in U$ is called the critical point if: $dF_p: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is not onto, and the image $F(p)$ is called Critical value. Otherwise, $c \in \mathbb{R}^m$ is called regular value.

Observation.

For a mapping $f: \mathbb{R}^3 \rightarrow \mathbb{R}$, the Jacobian at $p \in \mathbb{R}^3$ is

$$df_p = \begin{bmatrix} f_x(p) & f_y(p) & f_z(p) \end{bmatrix}$$

therefore, df_p is Not onto if and only if $f_x = f_y = f_z = 0$ at p .

This observation gives us that: for $f: U \subseteq \mathbb{R}^3 \rightarrow \mathbb{R}$,

$a \in f[U]$ is critical value if and only if f_x, f_y, f_z are not all zero on the preimage $f^{-1}[\{a\}]$.

[Thm.

If $a \in f[U]$ is regular value of a mapping $f: U(\subseteq \mathbb{R}^3) \rightarrow \mathbb{R}$,
then the preimage $f^{-1}[\{a\}] \subseteq \mathbb{R}^3$ is regular surface.

Proof. Let $p = (x_0, y_0, z_0) \in f^{-1}[\{a\}]$ be given. Since $a \in f[U]$ is regular value,
 f_x, f_y, f_z are not all zero. Without Loss of Generality, assume $f_z(p) \neq 0$.
Define $F: U \rightarrow \mathbb{R}^3: (x, y, z) \mapsto (x, y, f(x, y, z))$. Since f is differentiable, so is F .
And, the Jacobian of F at p is

$$dF_p = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ f_x & f_y & f_z \end{bmatrix}$$

is invertible, because $f_z(p) \neq 0$, therefore by the inverse function theorem,
for some neighbor V containing p and W containing $F(p)$,
 $F: V \rightarrow W$ is invertible and $F^{-1}: W \rightarrow V: (u, v, t) \mapsto (u, v, g(u, v, t))$ is differentiable.

Thm. Let $S \subseteq \mathbb{R}^3$ be a regular surface.

Given $p \in S$, there exists a neighbor V containing p such that

$V \cap S$ is a graph for some differentiable function either:
$$\begin{cases} Z = f(x, y) \\ Y = g(x, z) \\ X = h(y, z) \end{cases}$$

Proof. Let $X: \mathcal{U} \subseteq \mathbb{R}^2 \rightarrow S$ be a parametrization of S at p .

Denote $X(u, v) = (x(u, v), y(u, v), z(u, v))$. By the definition, at $q = X^{-1}(p) \in \mathcal{U}$,

$$DX_q = \begin{bmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \\ \frac{\partial z}{\partial u} & \frac{\partial z}{\partial v} \end{bmatrix} \text{ has rank } 2,$$

So, at least one of these is not zero at q :

$$\frac{\partial(x, y)}{\partial(u, v)}, \frac{\partial(y, z)}{\partial(u, v)}, \frac{\partial(x, z)}{\partial(u, v)}.$$

Assume $\frac{\partial(x, y)}{\partial(u, v)} \neq 0$. Consider a projection $\pi(x, y, z) = (x, y)$.

Then, $(\pi \circ X)(u, v) = (x(u, v), y(u, v))$.

By the assumption, $D(\pi \circ X)(q) \neq 0$, thus the inverse function theorem gives:

for some neighbor V_1 and V_2 containing q and $(\pi \circ X)(q)$, respectively,

$(\pi \circ X): V_1 \rightarrow V_2$ is diffeomorphism. Now, the inverse

$(\pi \circ X)^{-1}: V_2 \rightarrow V_1: (x, y) \mapsto (u(x, y), v(x, y))$ is differentiable, now,

$f: (x, y) \mapsto (x, y, z(u, v))$ is we desired. $\square$

Ellipsoid.

$$E = \left\{ (x, y, z) \in \mathbb{R}^3 \mid \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1 \right\}, \quad (a, b, c \neq 0)$$

Define $f: \mathbb{R}^3 \rightarrow \mathbb{R} : (x, y, z) \mapsto \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} - 1$

Then, $E = f^{-1}[\{0\}]$, f is differentiable because it is polynomial
and $0 \in f[E]$ is a regular point, because $f(x, y, z) = 0$ iff $(x, y, z) = \left(\frac{a}{\sqrt{3}}, \frac{b}{\sqrt{3}}, \frac{c}{\sqrt{3}}\right)$

but at this point, $df = (f_x \ f_y \ f_z) \neq 0$, so df is surjective.

Now, by the theorem, E is regular surface

Torus.

Consider a circle S' centered at $(0, a, 0)$, with a equation

$$(y-a)^2 + z^2 = r^2.$$

Rotate this with respect to z -axis, then we obtain a torus

$$z^2 = r^2 - (\sqrt{x^2 + y^2} - a)^2$$

That is, the torus is, for a map

$$f(x, y, z) = z^2 + (\sqrt{x^2 + y^2} - a)^2,$$

$$T = f^{-1}[\{r^2\}].$$

f is differentiable excepte $(x, y) \neq (0, 0)$, with

$$\frac{\partial f}{\partial x} = 2 \cdot (\sqrt{x^2 + y^2} - a)^2 \cdot \frac{x}{\sqrt{x^2 + y^2}}, \quad \frac{\partial f}{\partial y} = 2 \cdot (\sqrt{x^2 + y^2} - a)^2 \cdot \frac{y}{\sqrt{x^2 + y^2}},$$

$$\frac{\partial f}{\partial z} = 2z$$

[Sphere.

Consider a sphere S_+^2 centered at $(0,0,1)$, and $N=(0,0,2)$. ($x^2+y^2+(z-1)^2=1$)
Define a stereographic projection

$$\pi: S_+^2 \setminus \{N\} \rightarrow \mathbb{R}^2: (x,y,z) \mapsto \left(\frac{2x}{2-z}, \frac{2y}{2-z} \right)$$

1). π has an inverse

$$\pi^{-1}: \mathbb{R}^2 \rightarrow S_+^2 \setminus \{N\}: (u,v) \mapsto \left(\frac{4u}{u^2+v^2+4}, \frac{4v}{u^2+v^2+4}, \frac{2(u^2+v^2)}{u^2+v^2+4} \right)$$

It is deduced by: Consider a straight line

$$\ell(t) = (0,0,2) + t[(u,v,0) - (0,0,2)] = (tu, tv, 2-2t)$$

Find $\ell[\mathbb{R}] \cap S_+^2$, as

$$(tu)^2 + (tv)^2 + (2-2t)^2 = 1.$$

That is, $t^2u^2 + t^2v^2 + 4t^2 - 4t + 1 = 1 \Rightarrow t \cdot [t(u^2+v^2+4) - 4] = 0$

$$\text{so, } t=0 \quad \text{or} \quad t = \frac{4}{u^2+v^2+4}$$

Now, π, π^{-1} both are differentiable and 1-1, it is a parametrization.
Precisely, $\pi \circ \pi^{-1} = \text{id}$, so

$$D\pi \circ D\pi^{-1} = I_2 \Rightarrow D\pi^{-1} \text{ injective} \Rightarrow \text{rank } 2.$$

And, consider a Mongu Patch

$$f: D \rightarrow S_+^2: (x,y) \mapsto (x, y, 1 + \sqrt{1-x^2-y^2})$$

Then it covers the North $N \in S_+^2$, consequently, S_+^2 is regular surface